# Loss Functions: MSE and Cross-Entropy

Deep Learning

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# Contents of This Video

In this video, we will cover:

- What loss functions measure and why they matter
- Mean Squared Error (MSE) for regression
- Binary Cross-Entropy for classification
- Categorical Cross-Entropy for multi-class problems
- Choosing the right loss function

# What is a Loss Function?

**Loss = measure of prediction error**

$$L(y, \hat{y}) = \text{how wrong is } \hat{y} \text{ when truth is } y?$$

**Training = find weights that minimize average loss:**

$$\min_{W,b} \frac{1}{n} \sum_{i=1}^{n} L(y_i, \hat{y}_i)$$

**Key properties we want:**

- Loss = 0 when prediction is perfect
- Loss increases as prediction gets worse
- Differentiable (for gradient descent)

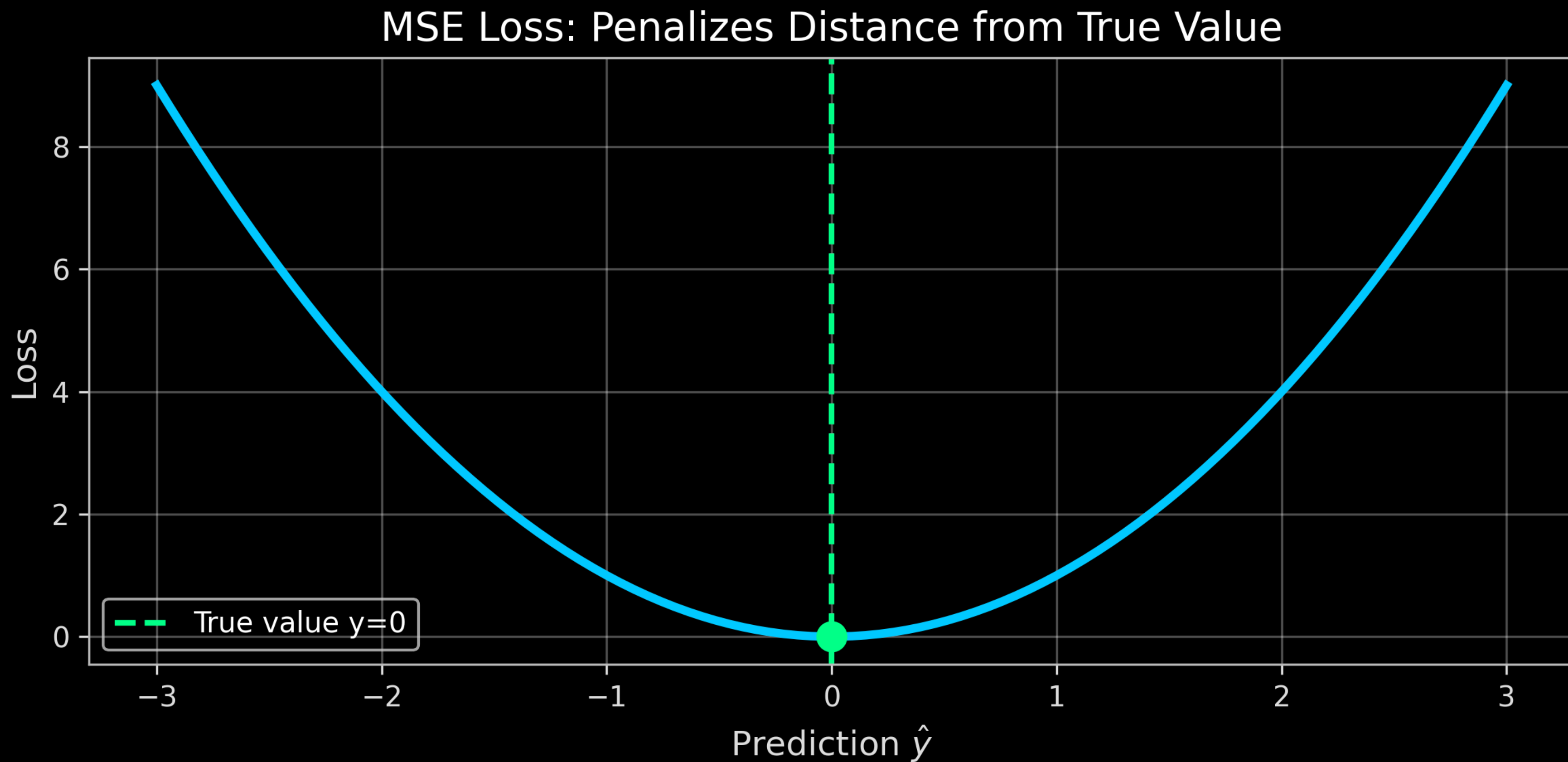
# Mean Squared Error (MSE)

**For regression tasks (predicting continuous values):**

$$L_{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y_i})^2$$

**For a single sample:**

$$L = (y - \hat{y})^2$$


MSE Loss: Penalizes Distance from True Value
Loss
0
2
4
6
8
−3
−2
−1
0
1
2
3
Prediction $\hat{y}$
True value y=0

# MSE Gradient

**The gradient tells us how to update predictions:**

$$\frac{\partial L}{\partial \hat{y}} = -2(y - \hat{y}) = 2(\hat{y} - y)$$

**Interpretation:**

- If $\hat{y} > y$: gradient is positive → decrease $\hat{y}$
- If $\hat{y} < y$: gradient is negative → increase $\hat{y}$
- Gradient magnitude proportional to error size

# Binary Cross-Entropy

**For binary classification (predicting probabilities):**

$$L_{BCE} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y_i}) + (1 - y_i)\log(1 - \hat{y_i})\right]$$

**For a single sample:**

$$L = -\left[y \log(\hat{y}) + (1 - y)\log(1 - \hat{y})\right]$$

**Where:**

- $y \in \{0, 1\}$ is the true label
- $\hat{y} \in (0, 1)$ is the predicted probability

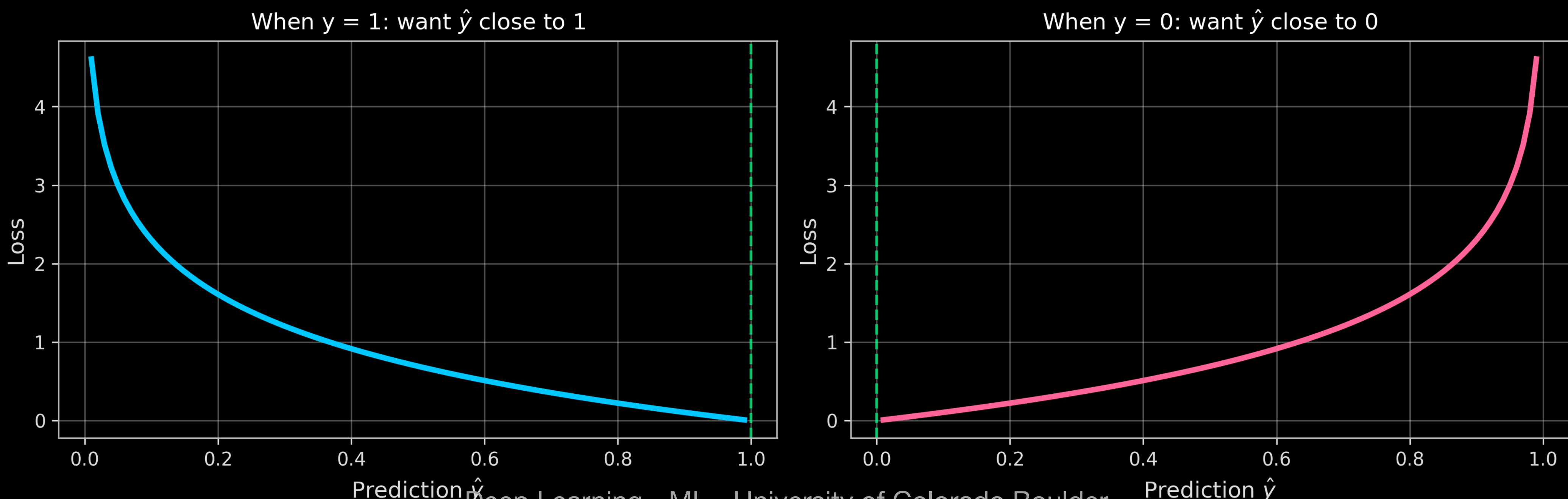
# Understanding Cross-Entropy

## Case 1: True label $y = 1$

$$L = -\log(\hat{y})$$

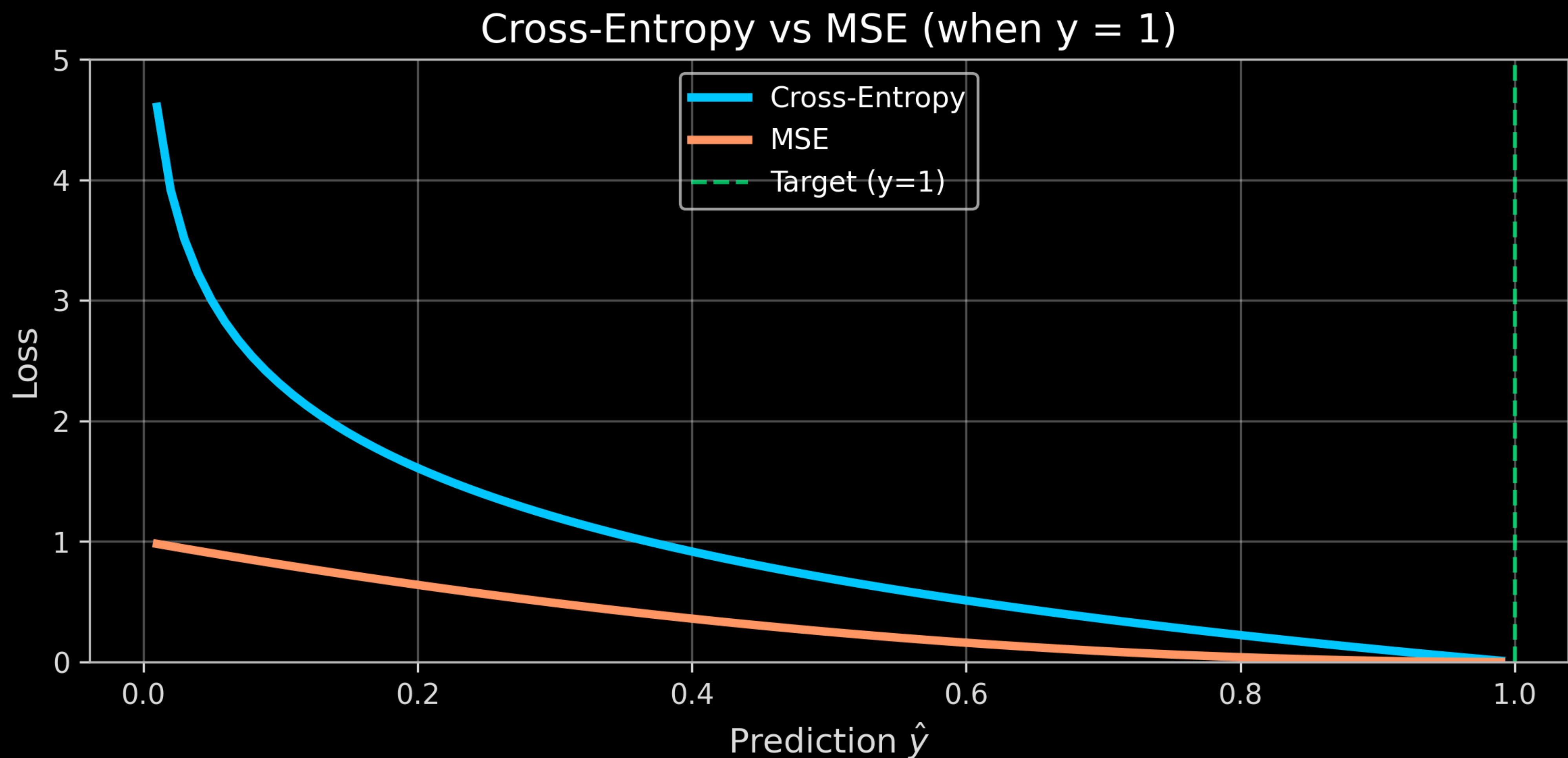
## Case 2: True label $y = 0$

$$L = -\log(1 - \hat{y})$$



# Why Cross-Entropy for Classification?

## Why not use MSE for classification?




**Cross-entropy has stronger gradients when predictions are very wrong!**

# Multi-Class: Categorical Cross-Entropy

**For $K$ classes with one-hot encoded labels:**

$$L = -\sum_{k=1}^{K} y_k \log(\hat{y_k})$$

**Example:** Classifying digits 0-9 ($K = 10$)

- True label for digit “3”: $y = [0,0,0,1,0,0,0,0,0,0]$
- Prediction: $\hat{y} = [0.01, 0.02, 0.05, 0.80, 0.03, \ldots]$
- Loss: $-1 \times \log(0.80) = 0.22$

# Softmax: Creating Probabilities for Multi-Class

**Output layer for multi-class: softmax activation**

$$\hat{y_k} = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

**Properties:**

- All outputs are positive: $\hat{y_k} > 0$
- Outputs sum to 1: $\sum_k \hat{y_k} = 1$
- Largest input gets largest probability

# Summary: Matching Loss to Task

| Task | Output Activation | Loss Function |
|---|---|---|
| Regression | None (linear) | MSE |
| Binary Classification | Sigmoid | Binary Cross-Entropy |
| Multi-Class Classification | Softmax | Categorical Cross-Entropy |

# What We've Covered

In this video, we've learned:

- Loss functions measure how wrong predictions are
- MSE for regression: $(y - \hat{y})^2$, penalizes large errors
- Binary Cross-Entropy: $-[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$
- Cross-entropy has stronger gradients for classification
- Softmax + Categorical Cross-Entropy for multi-class problems